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Approximating π : An Alternative First Week In Calculus I

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A class activity is described where students approximate π by estimating the circumference of the unit circle using inscribed polygons (Archimedes' method). This multi-day, group-based, interactive activity serves as an introduction to limits as well as a context to discuss some important mathematical ideas. The activity has been used by the author as the first portion of the first week of Calculus I as a partial replacement for standard Precalculus review.

0.1 Introduction

Standard review of Precalculus during the first week of Calculus I serves as a useful reminder of forgotten concepts, but does not display the beauty or challenge of Calculus. As an alternative to standard algebra and functions review I present an instructor-guided activity consisting of the following basic steps:

1. Students come to the idea (with some guidance) of approximating π through calculating perimeters of regular polygons inscribed in the unit circle.
2. Students calculate one such approximation of π .
3. Students use a general picture to find the relationship between the side-length of an inscribed polygon with the side-length of the inscribed polygon with twice the number of sides.
4. The instructor uses the relationship on a calculator and spreadsheet software to find successively better approximations to π .
5. The instructor introduces the picture, idea, and notation for limits at infinity.

This method of approximating π is certainly not the author's creation - it is essentially the method of Archimedes. However, I present here a structured method of guiding a class in this method (Section 0.3), with particular goals in mind (Section 0.2).

A suggestion for how to incorporate standard algebra and functions review in the midst of the activity and a possible structure for the classes immediately following are given in Section 0.4. The activity takes approximately two-and-a-half 50-minute class periods and has been used three times at a public university with class sizes of 25-30 students. At certain points a calculator is helpful for students, the instructor should have access to spreadsheet software in the classroom.

0.2 Objectives

The approximating π activity I describe was designed with a number of goals in mind:

1. Set a tone of group work and collaboration in the classroom. (The patterns created the first day and the first week of class guide how students participate from then on. I want my students working and interacting in groups from Day 1 so that group work is a natural and expected part of our class.)
2. Combine geometric and algebraic thinking. (Calculus consistently asks us to take something geometric and describe and analyze it algebraically, and conversely to take algebraic work and consider it's meaning geometrically. I want to start this translating as early as possible in the course.)
3. Have students construct an intimidating-looking formula that arises from a natural question. (Calculus is full of intimidating-looking formulas. I want to give my students a chance to create one of their own and get a sense that these formulas are natural and possible to be created, understood, and used.)
4. Introduce the idea of limits. (Moving from standard Precalculus review to limits always felt awkward to me. I want an activity that naturally leads to the core concept of limits.)
5. Give a sense of the challenging nature of Calculus. (Calculus is challenging and asks us to think in new ways, and I felt like that was hidden during standard Precalculus review. I want to give students an atypical and challenging experience to start off the semester.)
6. Remind and teach some important mathematical ideas. (For example, the fact that abstract math is free from the imperfections of physical measurement, approximating something we don't understand with something we do, choosing a parameter to simplify our considerations, important points and angles on the unit circle, the value of a strategy that makes use of previous knowledge to gain new knowledge, the value of a general picture to deduce a general formula, and the ability to work with multiple equations and variables.)

0.3 Description and Implementation

This section contains a summary of each day's goals followed by notes regarding details of implementation.

Day 1 (≈ 20 minutes)

Have the students answer the question “What is π ?” and think about “How could we approximate π ?”

Notes:

- I have the students think about “What is π ?” in small groups along with other ice breaker questions (their name, a TV show or movie they’ve been watching lately, and something about themselves), then ask them to share their “What is π ?” answers with the class.
 - I’m very hands-off with group formation (I just tell them to “circle up” into groups, letting them choose their own and stay at their desks), but I do limit group size to 4 or 5 students. There are of course other options: you could have students stand and work at a whiteboard together (instead of at desks), and if you wanted to randomize groups, you could verbally give each student a number (up through the number of groups you want), then have them find the others with their number. You could also choose each group manually if you’d like to.¹
 - Common answers to “What is π ?” are “a number, irrational, 3.14, 22/7, ratio of circumference to diameter” - I guide them to the fact that the reason we care about π is this last response, the fact that π captures a meaningful relationship in a circle.
 - Now is a good opportunity to “sell the students” on what they’ll start thinking about next (i.e. approximating π). I’ll appeal to their sense of independence and mention that the only reason they know 3.14 is because they were *told* that, and this is a chance to “build” something themselves, something that is theirs. You could also bring up power dynamics: “Imagine if your entire generation didn’t know why things were true. Those that did know why would hold a lot of power over you and could potentially abuse it. We don’t need to recreate everything, but knowing that we could is a powerful thought.” There are many different angles one could take to motivate the task - find one that resonates with you!
- I then pose “How could we approximate π ?” to the class and often the first response given is to physically measure the circumference and diameter of a circle.
 - I mention that with physical measurement, you won’t have a perfect circle and you won’t be able to measure that imperfect circle

¹For a discussion of best practices surrounding group work see for example, <https://uwaterloo.ca/centre-for-teaching-excellence/teaching-resources/teaching-tips/alternatives-lecturing/group-work/implementing-group-work-classroom>

exactly, but something beautiful about math is that we can work abstractly and precisely.

- At this point they generally don't have any other ideas. To help out I give them two nudges:
- First I ask “Is there anything we could choose to help our situation be more specific?” and they'll quickly come up with setting the radius or diameter. Setting $r = 1$ is common and is what I'll assume in the following, but the specific r doesn't matter (the geometric formulas in Days 2 and 3 will change if a different r is set). We now have $\pi = \frac{1}{2}C$, where C is the circumference of the unit circle, and our problem is reduced to estimating this circumference.
- The second nudge I give is that “curvy length is hard, but straight length is something we can compute if we know the endpoints” and I ask them to come up with an idea for how to approximate the circumference using straight lengths for the next class.

Day 2 (≈ 50 minutes)

Agree as a class how to approximate the circumference of the unit circle, compute one such approximation, think about how we could get a better but related approximation, and use the general picture for the doubled polygon to start writing relationships.

Notes:

- I start by giving them a few minutes to talk with their groups to share the thoughts they've had (or to come up with thoughts quickly).
 - Usually someone does come up with the idea of putting a regular polygon inside and finding the perimeter.
 - Our strategy is now $\pi = \frac{1}{2}C \approx \frac{1}{2}(\# \text{ of sides})(\text{length of a side})$ and I ask them to choose a letter to represent both of these variables. I also mention that since the length of a side depends on the number of sides it would be good to incorporate our first symbol in the second (for this paper I'll use n and l_n).
- I then ask them what simple polygon we could start with to find our first approximation to the circumference.
 - Usually a response of square or triangle is given (for this paper I'll assume triangle) and I'll draw the corresponding picture with a vertex at the point $(1, 0)$ (see Figure 1). We now need the coordinates of an adjacent vertex.

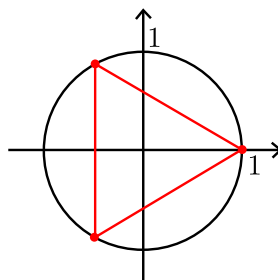


Figure 1: We use the perimeter of this triangle as our first approximation to the circumference of the unit circle.

- This is a good time to review some trigonometry, and afterwards I’ll give the coordinates in the form $(\cos(2\pi/3), \sin(2\pi/3)) = (-1/2, \sqrt{3}/2)$.
- I’ll then ask them to, in groups, find the length of the side and use it to get our first approximation to π (for the triangle we have $l_3 = \sqrt{3}$, and $\pi \approx 2.598$).
- I then ask them “How could we get better approximations?” and guide them to the idea of doubling the number of sides.
 - They will readily answer “More sides!” but the issue is that we don’t know what, e.g. $(\cos(2\pi/100), \sin(2\pi/100))$ is (and using our calculator is cheating since we’re trying to approximate π in the first place).
 - I’ll offer that “It would be great if we could find a better *but related* approximation” and ask what they think that could look like. They might come up with the idea of doubling the number of sides, but if they don’t I will describe it.
 - Our task now is to figure out the relationship between the side lengths so that we can use it to find better and better approximations of π .
- I’ll now introduce the general picture illustrating the relationship between an initial polygon and the polygon with doubled sides and have them start finding relationships.
 - I’ll say “Since our polygons are related there should be a way to take a *general picture* (where we don’t know n) and using relationships between the parts (e.g. the Pythagorean Theorem) come up with an equation telling us what l_{2n} is in terms of l_n .”
 - I’ll emphasize that for our general picture we’re not starting with our triangle - they will want to start with their first approximation.
 - I’ll draw a general picture similar to Figure 2.

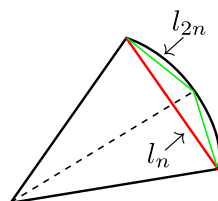


Figure 2: The general picture showing the relationship between the sides of a starting polygon and the polygon with doubled sides.

- I then ask them to, in groups, write down all the relationships they can and try to put them together to get our equation (I also give them time at the start of Day 3 to finish this).
- Some simple observations to help those that are stuck are that there are three lines in the picture that have length 1 (for the middle line, it's helpful to give names to the bigger and smaller pieces - if we use x and y then we have $x + y = 1$), and there are essentially two different right triangles in the picture and we can write what the Pythagorean Theorem says for each of them (in fact these are all the observations needed!).
- It may be helpful to tell them that the goal of our work is an equation with l_{2n} on one side and an expression on the other that will probably involve l_n .

Day 3 (≈ 50 minutes)

Have the students gather all the relationships from the general picture, build the main equation, use it to get the next approximation, use a spreadsheet to find many others, and visualize the approximations as a function and introduce limits at infinity.

Notes:

- After drawing the general picture and deciding as a class on common names for the larger and smaller parts of the middle line (in this paper I'll use x and y) I give the class about 10 minutes to work in groups and try to build our desired equation.
 - I then ask them for the key equations and ask them to tell me how to build our main equation.
 - The key equations are $x + y = 1$, $x^2 + (\frac{1}{2}l_n)^2 = 1$, and $y^2 + (\frac{1}{2}l_n)^2 = l_{2n}^2$ and the main equation is

$$l_{2n} = \sqrt{(l_n/2)^2 + (1 - \sqrt{1 - (l_n/2)^2})^2}.$$

- This is a great place to emphasize that this intimidating formula came from a very natural question and strategy.
- To get the next approximation (i.e. $n = 6$ if our initial approximation was with $n = 3$), I enter the main equation into my calculator on a document camera.
 - I invite the class to follow along and this is a good chance to remind them that parentheses are important when entering formulas.
 - After getting the next approximation, I emphasize that since our formula is general, we can keep using it as many times as we'd like and use spreadsheet software to speed the process.
- In a Google sheet I have the first approximation information saved, and columns for n , l_n (with our main equation already entered), and a column for our approximation of π using the formula $\pi \approx \frac{1}{2}nl_n$.
 - Dragging the main equation down the l_n column quickly gives side lengths for polygons with millions of sides which yield approximations of π that are accurate to all visible digits.
- I then tell them that we can think of this table of information as a function and graph it.
 - The function is $p(x)$, our x^{th} approximation of π , and I draw the function values over $x = 1$ (our triangle), 2, 3, and 4 as well as draw a dashed line at $y = \pi$ (see Figure 3).
 - The graph of $p(x)$ is discrete, but I encourage you to *not* mention this subtlety - it will most likely not bother students. Despite the discrete nature of this function, I prefer to use the variable x instead of n as I want the transition to functions such as $f(x) = 1/x$ to feel as smooth as possible.

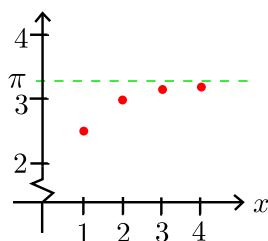


Figure 3: The graph of $p(x)$, whose outputs are our approximations of π

- As the big finale, I now lead them to the idea of a limit.
 - I like to ask in turn, “If we go out far enough, can we get within 0.00001 of π ?” “What if there were a thousand zeros before the 1,

could we still get that close?” “Could we get as close as we want?” They will quickly answer “Yes” to each of these questions.

- I then highlight their answer to the last question: “We can get as close as we want to π if we go out far enough, i.e. if x is big enough.” and say that because of this, we say that the limit of our function as $x \rightarrow \infty$ is π and we write $\lim_{x \rightarrow \infty} p(x) = \pi$.

0.4 After the Activity

Groupwork Precalculus Review (1 or 2 days)

Although the Approximating π activity naturally incorporates some review, a review of standard Precalculus topics is important. To facilitate this I use two large online homework assignments - one for each of algebra and functions - and ask the students to work on parts of them on the days that we are approximating π . On the day directly following the end of the activity, students bring in their laptops, cell phones, or a printout of the review assignments to work on the assignments in groups. In class I designate a side of the class for algebra review and a side for functions review. If a student finishes with both of those I also have a trig review online homework they can start on. The algebra and functions assignments are due at the end of the first weekend; the trigonometry review assignment is due at the end of the second weekend.

The Second Week of Class

Because of where the Approximating π activity finishes, I find it most natural to begin our standard class material with limits at infinity. I do not comment on the fact that we have moved from discrete functions (sequences) to continuous functions and students don't seem bothered by the switch. In computing limits at infinity algebraically, I end up using limit laws that I haven't proven yet, but again the students don't seem bothered.

After limits at infinity, I introduce the idea of finite limits (i.e. $\lim_{x \rightarrow a}$) and what they look like graphically, moving afterwards into the formal limit laws. I believe that this sequence sets up a strong intuitive foundation on which to then work algebraically. I do not assign any homework or give test questions based specifically on the Approximating π activity (I prefer to think of the activity more as a preparatory experience), but one could certainly do so.

0.5 Student Responses to Activity

I have not attempted an in-depth study of the efficacy of this approach; however an online survey of 24 of my students yielded encouraging results:

- When asked “Did you find the activity interesting?” 33% responded “Absolutely,” 58% responded “Yes,” 4% (or 1 student) responded “Not really,” and 4% responded “Not at all.”
- When asked “Did you find the activity challenging?” 21% responded “Absolutely,” 71% responded “Yes,” 8% responded “Not really,” and 0% responded “Not at all.”
- Here are a few selected responses to “What did you like about the activity?”
 - “It put π in a new perspective. In the past I’ve always seen it as 3.14 and now I have a visual representation of it.”
 - “The activities provided a nice intro to limits instead of just jumping in and writing down definitions.”
 - “I enjoyed the challenge and how it took several concepts to find the approximation. I also enjoyed how we worked in groups, as it helped to get to know one another.”
 - “I like the fact that one of the first problems we did had a really ugly answer especially since it was something that would start coming up a lot later.”

0.6 Conclusion

Approximating π has been a fun and challenging introduction to the world of Calculus. It leads from the familiar concepts of circles and π to the idea of limits at infinity in a stake-free, collaborative environment. Students are introduced to the idea of approximating a quantity and then making that approximation better, the power of a general relationship, and an intimidating-looking equation that came out of a natural question. The activity gives a sense of the challenge and beauty of Calculus and gives a strong reference point to use later in the semester when the “approximate and take a limit” strategy shows up in various forms.

Bibliography

- [1] Anna Anderegg, 2019, Thoughts.